AI-DRIVEN HEALTHCARE ANOMALY DETECTION SYSTEM

Submitted to

SMARTBRIDGE

PROJECT DOCUMENTATION

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Session

2025-26

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CHAPTER 1

ABSTRACT

Healthcare monitoring systems are becoming increasingly important due to the rising demand for continuous patient monitoring and early detection of medical abnormalities. Traditional monitoring systems often rely on manual observation or rule-based thresholds, which may fail to detect subtle or evolving anomalies in physiological data.

The AI-Driven Healthcare Anomaly Detection System aims to detect abnormal patterns in patient vital signs using machine learning and deep learning techniques combined with real-time data streaming technologies.

The system processes continuous streams of patient vitals such as heart rate, oxygen saturation (SpO_2), blood pressure, and body temperature. These values are analyzed using a hybrid anomaly detection approach combining an Autoencoder neural network and an Isolation Forest model.

To simulate real-world hospital monitoring systems, Apache Kafka is used to stream patient data in real time. A Kafka consumer performs machine learning inference on incoming data and detects anomalies. Detected anomalies are stored in a PostgreSQL database and visualized using a Flask-based web dashboard.

The system also includes an automated email alert mechanism that notifies healthcare professionals when critical anomalies are detected. By combining real-time streaming, machine learning, and interactive visualization, the system provides an efficient framework for proactive patient monitoring and early intervention.

CHAPTER 2

INTRODUCTION

Modern healthcare systems generate massive amounts of physiological data from monitoring devices such as heart rate monitors, pulse oximeters, and blood pressure sensors. Analyzing this data manually in real time is challenging and inefficient.

Artificial Intelligence (AI) and Machine Learning (ML) have opened new possibilities in healthcare analytics by enabling automated detection of abnormal patterns in patient data. These techniques can identify subtle variations in vital signs that may indicate potential health risks.

However, real-world healthcare monitoring systems must satisfy several requirements:

- Real-time data processing
- Scalable streaming infrastructure
- Reliable anomaly detection models
- Alerting mechanisms for medical staff
- Visualization dashboards for monitoring

This project integrates multiple modern technologies to create a real-time healthcare anomaly detection system capable of monitoring patient vitals continuously.

The system architecture includes:

- Apache Kafka for real-time streaming
- Machine Learning models for anomaly detection
- PostgreSQL database for anomaly storage
- Flask backend for API services
- Web dashboard for visualization
- Email notification system for alerts

The solution demonstrates how AI and streaming technologies can be combined to build intelligent healthcare monitoring platforms.

CHAPTER 3

PROBLEM STATEMENT

Healthcare monitoring systems play a crucial role in ensuring patient safety and timely medical intervention. Modern hospitals rely on monitoring devices that continuously record physiological parameters such as **heart rate, oxygen saturation (SpO_2), blood pressure, and body temperature**. These vital signs provide important insights into the overall health condition of a patient and help medical professionals detect potential health risks.

However, the continuous generation of patient data creates several challenges in terms of analysis, monitoring, and timely detection of abnormal conditions. The increasing volume of healthcare data makes it difficult for healthcare professionals to manually analyze and monitor every patient's physiological parameters in real time. As a result, there is a growing need for intelligent systems that can automatically detect anomalies in patient vital data and assist medical professionals in decision-making.

The major problems associated with traditional healthcare monitoring systems are discussed below.

1. Difficulty in Monitoring Large Volumes of Patient Data

Modern healthcare environments generate a massive amount of physiological data through monitoring devices. Each patient may produce continuous streams of vital readings throughout the day. When multiple patients are being monitored simultaneously, the volume of data becomes extremely large.

Because of this:

- Healthcare professionals may find it difficult to continuously observe every parameter.
- Important abnormal patterns in patient data may go unnoticed.
- Manual monitoring increases workload and reduces efficiency.

Therefore, an automated system capable of analyzing large volumes of patient data is required.

2. Limitations of Threshold-Based Monitoring Systems

Many existing monitoring systems rely on **fixed threshold values** to trigger alerts. For example, an alert may be generated if the heart rate exceeds a predefined limit or if oxygen saturation drops below a certain level.

However, this approach has several limitations:

- It cannot detect **subtle or gradual changes** in vital signs.
- Important patterns that remain within threshold limits may still indicate health risks.
- Threshold values may not be suitable for every patient due to individual differences.

As a result, threshold-based monitoring systems often fail to detect complex abnormalities in patient health conditions.

3. Challenges in Detecting Temporal Patterns in Vital Signs

Patient physiological data is **time-dependent**, meaning that the interpretation of a reading often depends on previous measurements. For example, a gradual increase in heart rate or a continuous decrease in oxygen saturation over time may indicate a deteriorating health condition.

Traditional monitoring methods usually analyze individual readings instead of considering the overall pattern of change.

This creates several issues:

- Important trends in patient vitals may not be identified.
- Temporary fluctuations may trigger false alarms.
- Sustained anomalies may remain undetected.

Therefore, it is important to develop systems that can analyze **sequential patterns in patient data**.

4. Lack of Real-Time Intelligent Monitoring Systems

Another major problem is the lack of systems that can **analyze patient data in real time using intelligent algorithms**. In critical healthcare situations, delays in detecting abnormal health conditions can lead to serious medical complications.

Without automated systems:

- Healthcare professionals must manually interpret monitoring data.
- Critical abnormalities may not be detected immediately.
- Early warning signs of medical emergencies may be missed.

This highlights the need for intelligent healthcare systems capable of performing real-time analysis and anomaly detection.

5. Absence of Integrated Monitoring and Alerting Systems

Even when anomalies are detected, many healthcare monitoring systems lack proper mechanisms for **visualizing and communicating abnormal conditions effectively**.

Some common issues include:

- Limited visualization tools for monitoring patient trends.
- Lack of automated alerts for critical anomalies.
- Difficulty in accessing historical monitoring data.

An effective healthcare monitoring solution should therefore provide **data visualization, anomaly alerts, and historical data storage** in an integrated platform.

Problem Addressed by the Proposed System

To address the challenges mentioned above, this project proposes an **AI-driven healthcare anomaly detection system** that can automatically analyze patient vital data and detect abnormal patterns.

The system aims to solve the identified problems by:

- Using **machine learning models** to detect complex anomalies in physiological data.
- Applying **temporal analysis techniques** to capture patterns over time.
- Integrating **Apache Kafka for real-time data streaming**.
- Providing **interactive dashboards for monitoring patient vitals**.
- Implementing **automated alert systems for critical anomalies**.

By combining artificial intelligence with real-time data processing and visualization technologies, the proposed system aims to create a more efficient and reliable healthcare monitoring framework that supports early detection of potential medical risks.

CHAPTER 4

OBJECTIVES OF THE PROJECT

The main objective of this project is to develop an AI-driven healthcare anomaly detection system that can automatically monitor patient vital signs and detect abnormal health patterns. By combining machine learning models, real-time data streaming, and visualization tools, the system aims to provide an efficient framework for continuous patient monitoring and early detection of potential health risks.

The key objectives of the project are as follows:

1. To Develop an Intelligent Anomaly Detection System

The primary objective of the system is to design an intelligent framework capable of identifying abnormal patterns in patient vital signs. By applying machine learning techniques to physiological data such as heart rate, oxygen saturation (SpO_2), blood pressure, and body temperature, the system can distinguish between normal and abnormal conditions. This helps in detecting potential health issues at an early stage and supports better medical decision-making.

2. To Capture Temporal Patterns in Patient Vital Data

Patient health data is sequential in nature and changes over time. Therefore, the system implements a sliding window temporal modeling approach that analyzes sequences of vital readings rather than individual measurements. This allows the system to identify sustained abnormalities and reduces false detections caused by temporary fluctuations or sensor noise.

3. To Implement Hybrid Machine Learning Models

Another important objective is to improve anomaly detection accuracy by combining multiple machine learning techniques. The system integrates an Autoencoder deep learning model and an Isolation Forest algorithm. The Autoencoder learns the normal behavior of patient vitals and detects anomalies through reconstruction error, while Isolation Forest identifies statistical outliers in the dataset. The combination of these models enhances the reliability of anomaly detection.

4. To Enable Real-Time Data Processing

The project aims to simulate real-time patient monitoring by integrating Apache Kafka as a data streaming platform. Kafka enables continuous ingestion and processing of patient vital data. A Kafka producer generates simulated patient readings, while a Kafka consumer processes the incoming data and performs real-time anomaly detection using trained machine learning models.

5. To Classify Detected Anomalies Based on Severity

The system also aims to categorize detected anomalies according to their severity levels. Based on the anomaly score generated by the machine learning models, abnormalities are classified into LOW, MEDIUM, and HIGH severity levels. This classification helps healthcare professionals prioritize cases that require immediate medical attention.

6. To Provide Visualization and Automated Alerts

Another objective of the project is to provide a web-based monitoring dashboard that displays real-time graphs of patient vitals and anomaly alerts. In addition, the system includes an automated email alert mechanism that notifies healthcare providers when high-severity anomalies are detected. This ensures that critical situations are identified quickly and appropriate medical action can be taken.

CHAPTER 5

DATASET DESCRIPTION

The system uses a **simulated healthcare vitals dataset** designed to represent real-world physiological monitoring data. Due to strict privacy regulations and limited availability of real hospital datasets, simulated datasets are commonly used in research and development of healthcare monitoring systems.

The dataset mimics real-time patient monitoring by generating realistic vital sign readings over time.

Each record represents a snapshot of a patient's physiological condition at a particular timestamp.

Features Used

The dataset includes the following patient vital parameters:

Feature	Description
Patient ID	Unique identifier for each patient
Heart Rate (BPM)	Number of heartbeats per minute
SpO_2 (%)	Blood oxygen saturation level
Blood Pressure (mmHg)	Systolic/Diastolic pressure
Body Temperature (°C)	Patient body temperature
Timestamp	Time when vitals were recorded

These physiological parameters are widely used in clinical monitoring systems to assess patient health conditions.

Dataset Characteristics

The simulated dataset is designed to replicate realistic healthcare monitoring scenarios.

Characteristics include:

- Continuous time-series vital measurements
- Normal physiological ranges for most observations
- Injected abnormal values to simulate medical anomalies
- Multiple patient records for monitoring scenarios

The dataset is used to train and evaluate anomaly detection models capable of identifying unusual patterns in patient health data.

Why Simulated Data is Used

Using simulated healthcare data provides several advantages:

- Protects **patient privacy**
- Allows **controlled anomaly generation**
- Enables **testing of real-time monitoring systems**
- Provides flexibility in creating different medical scenarios

This makes simulated datasets ideal for developing and validating AI-driven healthcare monitoring systems.

CHAPTER 6

SYSTEM ARCHITECTURE AND REAL-TIME STREAMING

The overall system architecture is designed to enable **continuous monitoring of patient vital signs and real-time anomaly detection**. The architecture integrates machine learning models, streaming infrastructure, database storage, and a visualization dashboard into a unified pipeline.

The workflow of the system begins with the **Patient Vital Dataset**, which contains physiological parameters such as heart rate, oxygen saturation (SpO_2), blood pressure, and body temperature. These parameters represent key indicators of a patient's health condition and are used to train and evaluate the anomaly detection models.

System Workflow

The major stages of the system architecture are summarized below:

1. **Patient Vital Dataset**
    - Contains physiological measurements collected from patient monitoring systems.
    - Includes parameters such as heart rate, SpO_2, blood pressure, and temperature.
2. **Data Preprocessing**
    - Cleaning and formatting of raw healthcare data.
    - Handling missing values and removing inconsistent records.
    - Selecting relevant features required for anomaly detection.
3. **Sliding Window Temporal Modeling**
    - Healthcare data is sequential and time-dependent.
    - A sliding window technique is used to group consecutive readings together.

- This approach helps detect **sustained anomalies instead of isolated noisy readings**.

4. **Machine Learning Models (Autoencoder + Isolation Forest)**
   The anomaly detection engine combines two complementary models:

- **Autoencoder (Deep Learning Model)**
  - Learns the normal patterns of physiological data.
  - Detects anomalies using reconstruction error.
- **Isolation Forest (Machine Learning Model)**
  - Identifies statistical outliers in the dataset.
  - Efficient for high-dimensional anomaly detection.

5. **Anomaly Score Calculation**
   - Each model generates an anomaly score.
   - These scores are combined to produce a **final risk score** for each observation.
6. **Severity Classification**
   Based on the final risk score, anomalies are categorized into:
   - **LOW** – Minor deviation from normal vital ranges
   - **MEDIUM** – Moderate abnormality requiring monitoring
   - **HIGH** – Critical condition requiring immediate attention

Real-Time Data Streaming using Apache Kafka

To simulate a real-world hospital monitoring system, the project uses **Apache Kafka**, which enables high-throughput real-time data streaming.

The streaming architecture consists of the following components:

Kafka Producer

The Kafka Producer simulates continuous patient vital readings.

Functions of the producer include:

- Generating simulated patient vital data
- Sending records to a Kafka topic
- Publishing new readings at regular time intervals

This component mimics **real medical monitoring devices**.

Kafka Consumer

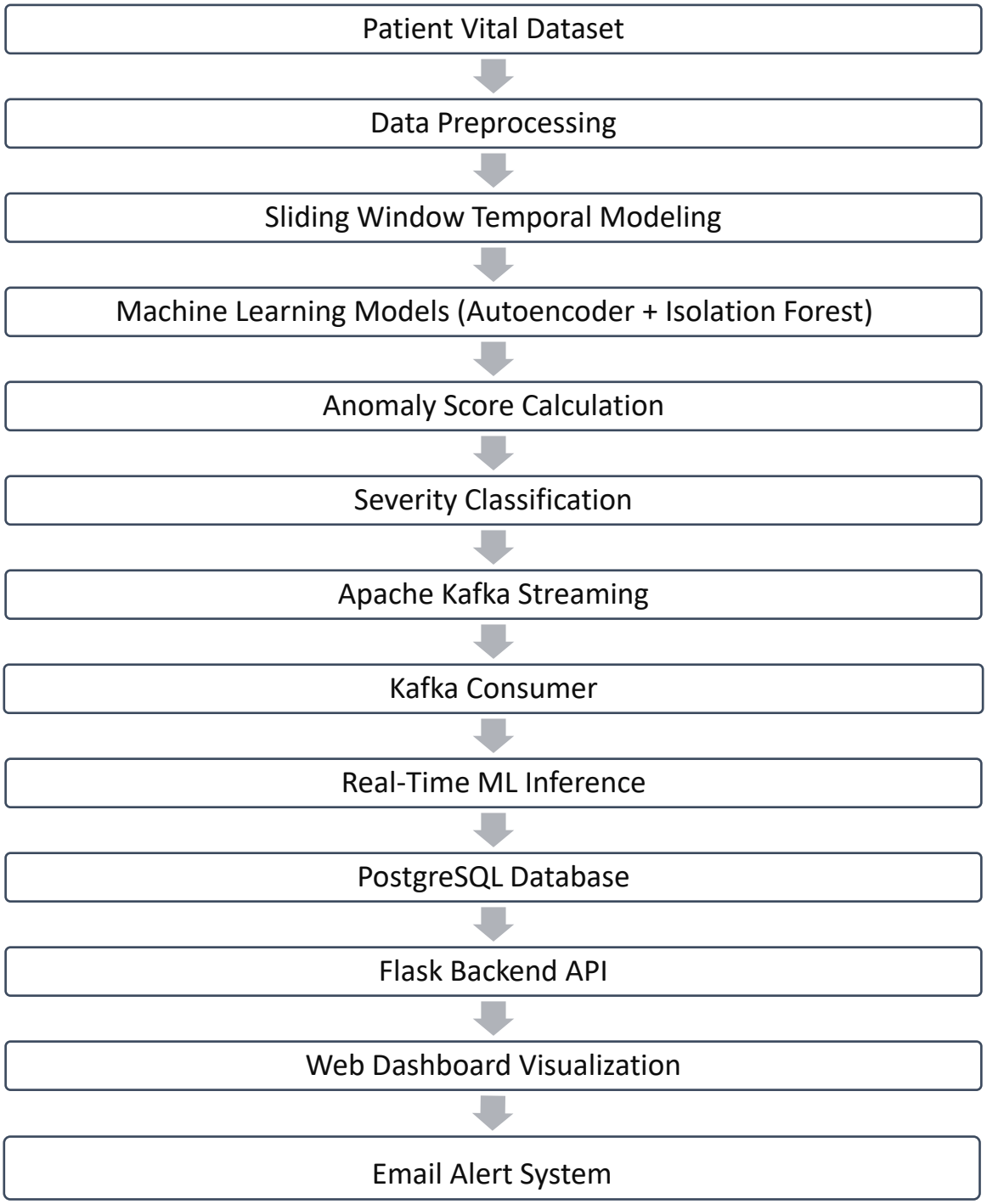
The Kafka Consumer retrieves the streaming data from the Kafka topic and performs real-time analysis.

Key responsibilities of the consumer include:

- Receiving live patient vitals from Kafka
- Applying preprocessing and feature scaling
- Running trained anomaly detection models
- Generating anomaly scores and severity levels

This allows the system to **detect abnormal health conditions in real time**.

The architecture of the AI-Driven Healthcare Anomaly Detection System integrates machine learning, streaming, database management, and visualization components.



CHAPTER 7

MACHINE LEARNING MODELS

The anomaly detection framework combines two complementary models:

1. Autoencoder (Deep Learning Model)

An Autoencoder is a neural network designed to reconstruct input data. It learns the normal patterns of patient vitals during training.

When abnormal data is encountered, the reconstruction error becomes high, indicating a potential anomaly.

The architecture of the Autoencoder used in this project includes multiple dense layers that compress and reconstruct the input feature space.

(Your model architecture image fits perfectly here in the documentation.)

2. Isolation Forest

Isolation Forest is an unsupervised anomaly detection algorithm that isolates anomalies by randomly partitioning data.

Key characteristics:

- Works efficiently on high-dimensional data
- Detects outliers based on isolation depth
- Suitable for real-time anomaly detection

Hybrid Anomaly Detection

Anomaly scores from both models are combined to produce a final risk score.

Based on this score, severity levels are classified into:

- LOW – Minor deviations
- MEDIUM – Moderate anomalies
- HIGH – Critical anomalies

CHAPTER 8

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand the characteristics, distribution, and relationships of patient vital signs before training machine learning models. Proper analysis helps identify trends, outliers, and data inconsistencies.

The patient vital signs dataset includes physiological parameters such as **Heart Rate, SpO_2 (oxygen saturation), Blood Pressure, and Body Temperature**. These parameters are commonly used in healthcare monitoring systems to assess patient health conditions.

Objectives of EDA

The key objectives of exploratory data analysis in this project are:

- Understanding distribution of vital parameters
- Identifying abnormal values and outliers
- Checking correlations between physiological parameters
- Ensuring data quality before model training

Data Distribution Analysis

Histograms and statistical summaries were used to examine how patient vitals are distributed across the dataset.

Typical healthy ranges include:

Vital Sign	Normal Range
Heart Rate	60 – 100 BPM
SpO_2	95 – 100 %
Body Temperature	36.1°C – 37.2°C
Blood Pressure	~120/80 mmHg

Values significantly outside these ranges may indicate abnormal physiological conditions.

Correlation Analysis

Correlation analysis was performed to understand relationships between features. For example:

- Low SpO_2 may correlate with abnormal heart rate
- High temperature may correlate with elevated heart rate
- Blood pressure changes may indicate cardiovascular stress

Understanding these relationships helps machine learning models learn **normal physiological patterns** more effectively.

CHAPTER 9

BACKEND SYSTEM, DATABASE AND DASHBOARD

The backend of the healthcare monitoring system is implemented using Flask, a lightweight Python web framework used for building RESTful APIs and backend services. Flask acts as the communication layer between the anomaly detection system, the database, and the web-based monitoring dashboard.

The backend is responsible for retrieving anomaly records from the database and providing them to the dashboard through API endpoints. These APIs allow the frontend interface to fetch real-time monitoring data dynamically.

For persistent data storage, the system uses a PostgreSQL database. PostgreSQL stores detected anomalies along with associated patient information and physiological measurements. Each anomaly record typically includes fields such as patient ID, heart rate, SpO_2 value, body temperature, blood pressure, anomaly score, severity level, and timestamp.

Storing anomaly data in a relational database enables historical analysis and patient-wise monitoring. Healthcare professionals can review past anomalies, observe trends in patient vitals, and identify long-term health patterns.

The system also includes a web-based monitoring dashboard developed using HTML, CSS, JavaScript, and Chart.js. The dashboard provides an intuitive visual interface for monitoring patient health conditions in real time.

Several visual components are included in the dashboard:

- Line charts displaying heart rate trends
- SpO_2 monitoring graphs
- Real-time anomaly alerts
- Severity level indicators
- Patient monitoring summaries

These visualizations help healthcare professionals quickly understand patient conditions and respond appropriately.

CHAPTER 10

ALERT SYSTEM, TESTING AND DEPLOYMENT

Early detection of critical anomalies is extremely important in healthcare monitoring systems. To ensure rapid response, the system includes an automated email alert mechanism.

When a high-severity anomaly is detected by the anomaly detection models, the system automatically generates an alert message containing relevant patient information. This message is sent to healthcare providers via email using SMTP-based Gmail integration with App Password authentication.

The alert email typically contains the patient ID, vital readings, anomaly severity level, and the timestamp at which the anomaly occurred. This mechanism ensures that healthcare professionals are immediately notified about potentially dangerous health conditions.

Before deployment, the system undergoes multiple stages of testing to ensure reliability and proper functioning of all components. Unit testing is performed on individual modules such as model prediction functions, Kafka consumers, and API endpoints. Integration testing is then conducted to verify that all system components work together seamlessly.

Real-time simulation testing is also performed by running Kafka producers that continuously generate patient vital readings. These simulated streams allow the system to be tested under conditions similar to real hospital monitoring environments.

Deployment of the system involves installing required Python dependencies, starting Kafka and Zookeeper services, launching the Kafka producer and consumer processes, and running the Flask backend server. Once deployed, the monitoring dashboard can be accessed through a web browser.

CHAPTER 11

CONCLUSION

The AI-Driven Healthcare Anomaly Detection System demonstrates how machine learning and real-time streaming technologies can be combined to create an intelligent healthcare monitoring solution. By analyzing patient vital signs continuously and detecting abnormal patterns automatically, the system provides a powerful framework for early detection of medical risks.

The integration of **Autoencoder-based deep learning models, Isolation Forest anomaly detection, Apache Kafka streaming infrastructure, PostgreSQL database storage, and interactive dashboards** enables efficient monitoring of patient health conditions in real time.

Such intelligent monitoring systems can significantly improve healthcare outcomes by assisting medical professionals in identifying abnormal physiological conditions early and enabling timely medical intervention.